

Research Conclusions

Europe's AI boom is not inherently environmentally sustainable. Its sustainability depends primarily on each country's renewable-energy mix and digital infrastructure.

Key Findings

- **Enterprise AI adoption alone is not associated with higher CO₂ emissions.** Countries with higher AI adoption do not necessarily produce more emissions.
- **Data-centre infrastructure shows the strongest relationship with CO₂ emissions.** Countries with more listed data centres generally have higher power-sector emissions, although this is also influenced by country size, electricity demand, industrial activity, and digital-hub status.
- **Renewable energy plays a key role in reducing environmental impact.** Countries with stronger renewable-energy systems generally experience lower emissions and are better positioned for sustainable AI growth.
- **The most sustainable conditions combine AI adoption with high renewable-energy shares.** In contrast, countries with expanding digital infrastructure but weaker renewable-energy systems may face greater environmental pressure.
- **Correlation does not imply causation.** This analysis identifies country-level patterns but does not prove that AI or data centres directly cause higher emissions.

Final Answer

Europe's AI boom can be environmentally sustainable, but only when digital infrastructure expands alongside renewable energy and cleaner electricity systems.

Next Steps

The main area for further research is the relationship between data-centre infrastructure and CO₂ emissions. Although countries with more listed data centres tended to have higher power-sector emissions in this analysis, other national factors may be influencing this relationship.

A stronger follow-up analysis should control for factors such as population, GDP, industrial activity, total electricity demand, and national electricity generation. It should also use data-centre capacity or electricity consumption, where available, rather than only the number of listed facilities. Finally, historical data-centre records would allow the relationship to be studied over time instead of using a single current infrastructure snapshot.

These steps would help determine whether the observed relationship remains meaningful after accounting for broader country-level differences.